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2. What is the displacement amplitude of sound with frequency $f = 1.5$ kHz and intensity level $\beta = 90$ dB. Take the speed of sound to be $v = 340$ m/s and the density of air to be $\rho = 1.3$ kg/m^3. (Answer 0.23 μm)

blood cells. When the reflected sound returns to the probe it is combined with the emitted sound and fed to a speaker. The speaker produces audible sound at the beat frequency. The technician can listen to the speaker to better position the probe. If the technician holds the probe at an angle such that the blood is flowing directly away from the probe the speaker produces sound at $f_b = 200$ Hz. Using this information and the fact that the speed of sound in blood is $v = 1500$ m/s, what is the velocity of the blood flow? (Answer 6.0 m/s)

Doppler Effect: $f' = f\frac{V+V_D}{V+V_S}$

- here, the first shift happens as the transmitted wave of blood flow:

$$f_1 = f_0 \frac{V+V_{blood}}{V}$$

↳ source (probe) is not moving, so $V_S = 0$

↳ blood is the detector moving away from probe

- second shift happens as the reflected wave of blood flow:

$$f_2 = f_1 \frac{V}{V-V_{blood}}$$

↳ source is the blood getting reflected

↳ receiver (probe) is stationary

- so $f_2 = f_0 \frac{V+V_{blood}}{V-V_{blood}}$

- $f_{beat} = f_2 - f_0$

$$f_b = f_0 \frac{V+V_{blood}}{V-V_{blood}} - f_0 = f_0\left(1 - \frac{V+V_{blood}}{V-V_{blood}}\right)$$

$$= f_0\left(\frac{V-V_{blood} - V - V_{blood}}{V-V_{blood}}\right)$$

$$= f_0\left(\frac{-2V_{blood}}{V-V_{blood}}\right)$$

$$\cong 2f_0 \frac{V_{blood}}{V}, \text{ since } V_{blood} << V$$

- $V_{blood} = \frac{f_b V}{2f_0}$

$$= \frac{(200\text{ Hz})(1500\text{ m/s})}{2(25000\text{ Hz})} = \boxed{6\text{ m/s}}$$

2. What is the displacement amplitude of sound with frequency $f = 1.5$ kHz and intensity level $\beta = 90$ dB. Take the speed of sound to be $v = 340$ m/s and the density of air to be $\rho = 1.3$ kg/m^3. (Answer 0.23 μm)

given:
$$\begin{cases} f = 1.5 \text{ kHz} \\ \beta = 90 \text{ dB} \\ v = 340 \frac{m}{s} \\ \rho = 1.3 \frac{kg}{m^3} \end{cases}$$

- $I = \frac{1}{2} \rho v \omega^2 s_m^2$
- $\beta = (10\,dB) \log\left(\frac{I}{I_0}\right)$

  ↳ $90\,dB = 10\,dB \cdot \log\left(\frac{I}{I_0}\right)$

  ↳ $9 = \log\left(\frac{I}{I_0}\right)$

  $10^9 = \frac{I}{I_0} = \frac{I}{10^{-12}}$

  $I = 10^9 \cdot 10^{12} = 10^{-3} \frac{W}{m^2}$
- $\frac{\omega}{2\pi} = f \rightarrow \omega = 2\pi f = 2\pi (1.5\,kHz) = 9.42\,kHz$
- $10^{-3} \frac{W}{m^2} = \frac{1}{2}\left(1.3 \frac{kg}{m^3}\right)\left(340 \frac{m}{s}\right)(9.42\,kHz)^2 s_m^2$

  $= \left(\frac{1}{2} \times 1.3 \times 340 \times 9.42^2 \times 10^{-6}\right)\left(\frac{kg}{m^3} \cdot \frac{m}{s} \cdot \frac{1}{s^2}\right) s_m^2$

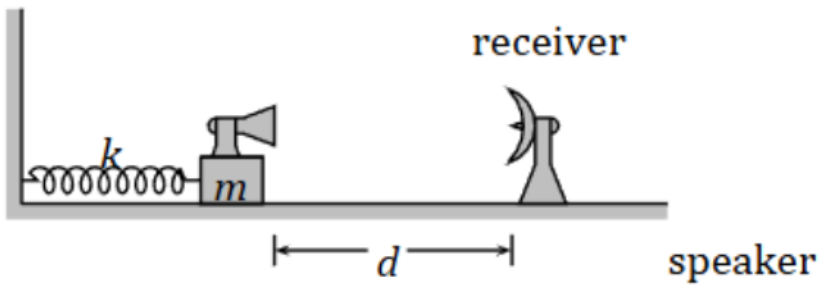
  $= 1.96 \times 10^{-2} \frac{kg}{m^2 s^3} s_m^2$

↓ $10^{-3} \frac{kg\,m^2}{s^3} = 1.96 \times 10^{-2} \frac{kg}{s^3} s_m^2$

$s_m^2 = \frac{1}{1.96} \times 10^{-1} m^2$

$\boxed{s_m = 0.23\,\mu m}$

3 .In the picture below, the total mass of the speaker and the block it is attached to is m = 0.50 kg. The spring has a force constant k = 3200 N/m. The speaker emits a tone at a frequency f = 8.0 kHz. When the spring is at its equilibrium length the distance between the speaker and receiver is d = 2.0 m. Also, when at equilibrium, the intensity level of the tone at the receiver is β = 40 dB. The block is pulled a distance A = 0.50 m to the right and released. The block and attached speaker begin moving back and forth in simple harmonic motion. Take the speed of sound to be v = 340 m/s



a. Determine the range of the tone intensity level heard by the receiver. (Answer 4.4 dB)
b. Determine the range of the tone frequency heard by the receiver. (Answer 1.9 kHz)

a) $r_1 = 2m$, $r_2 = 1.5m$

$$I_2 = \frac{\text{power}}{4\pi r_2^2}$$

$$\frac{I_2}{I_1} = \frac{\frac{\text{power}}{4\pi r_2^2}}{\frac{\text{power}}{4\pi r_1^2}} = \frac{r_1^2}{r_2^2} = \frac{(2m)^2}{(1.5m)^2} = 1.778$$

$$I_2 = 1.778 \times 10^{-8} \frac{W}{m^2}$$

$$\frac{I_3}{I_1} = \frac{r_1^2}{r_3^2} = \frac{(2m)^2}{(2.5m)^2} = 0.64$$

$$I_3 = 0.64 \times 10^{-8} \frac{W}{m^2}$$

$$\beta_2 = 10\log\left(\frac{1.778\times10^{-8}}{10^{-12}}\right) = 42.5dB$$

$$\beta_3 = 10\log\left(\frac{0.64\times10^{-8}}{10^{-12}}\right) = 38.1\ dB$$

$$\text{range} = \beta_2 - \beta_3 = \boxed{4.4dB}$$

b) $x(t) = A\cos(\omega t)$

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{3200}{0.5}} = 80\ \frac{1}{s}$$

$$f_L = \frac{v}{v+v_s} f_s = \frac{340\frac{m}{s}}{(340+40)\frac{m}{s}}(8\ kHz) = 7.16\ kHz$$

$$f_L = \frac{v}{v-v_s} f_s = \frac{340}{(340-40)}(8kHz) = 9.07\ kHz$$

$$(9.07 - 7.16)kHz = \boxed{1.9kHz}$$